

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

**BISMILLAH ARRAHMAN ARRAHEEM**

# Artificial Intelligence (CS-401)

## Linear Regression

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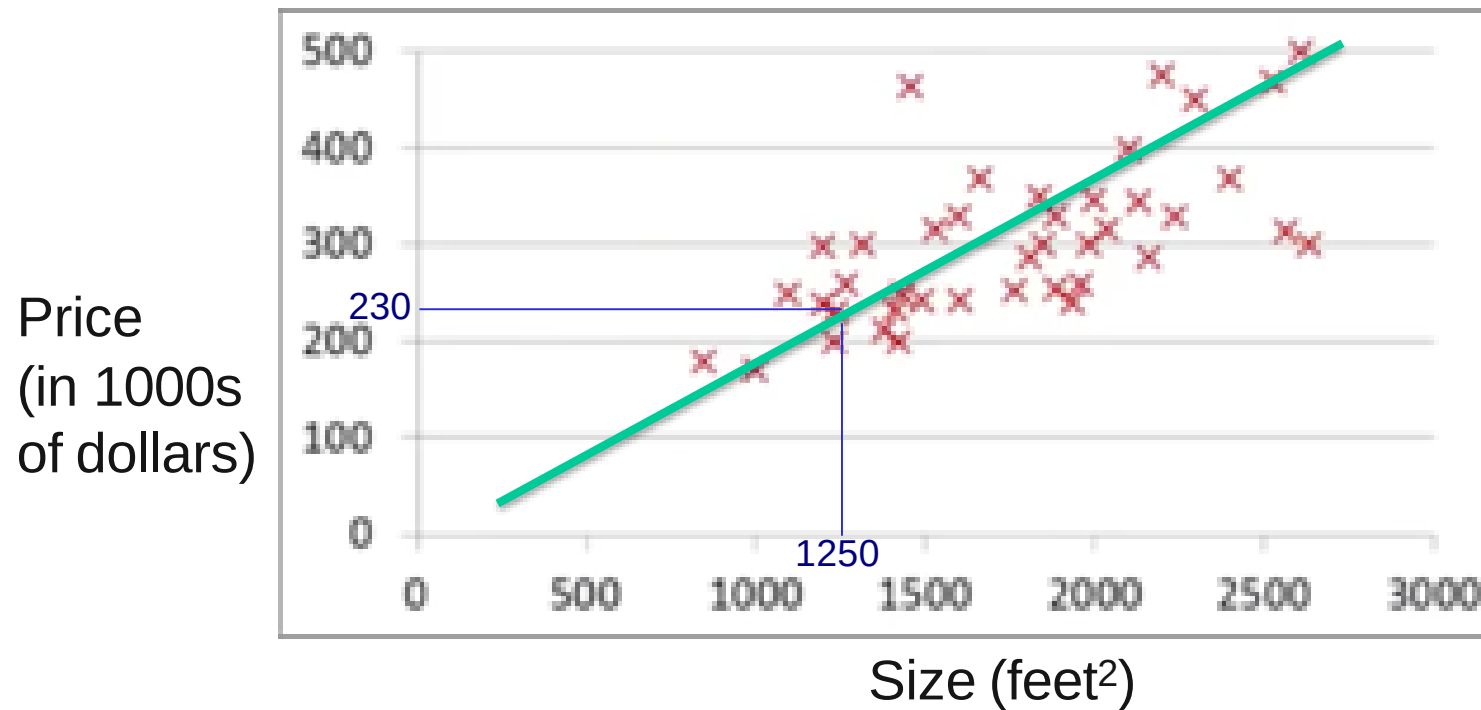
PhD in Artificial Intelligence

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# Motivation

Given the following **housing prices**,



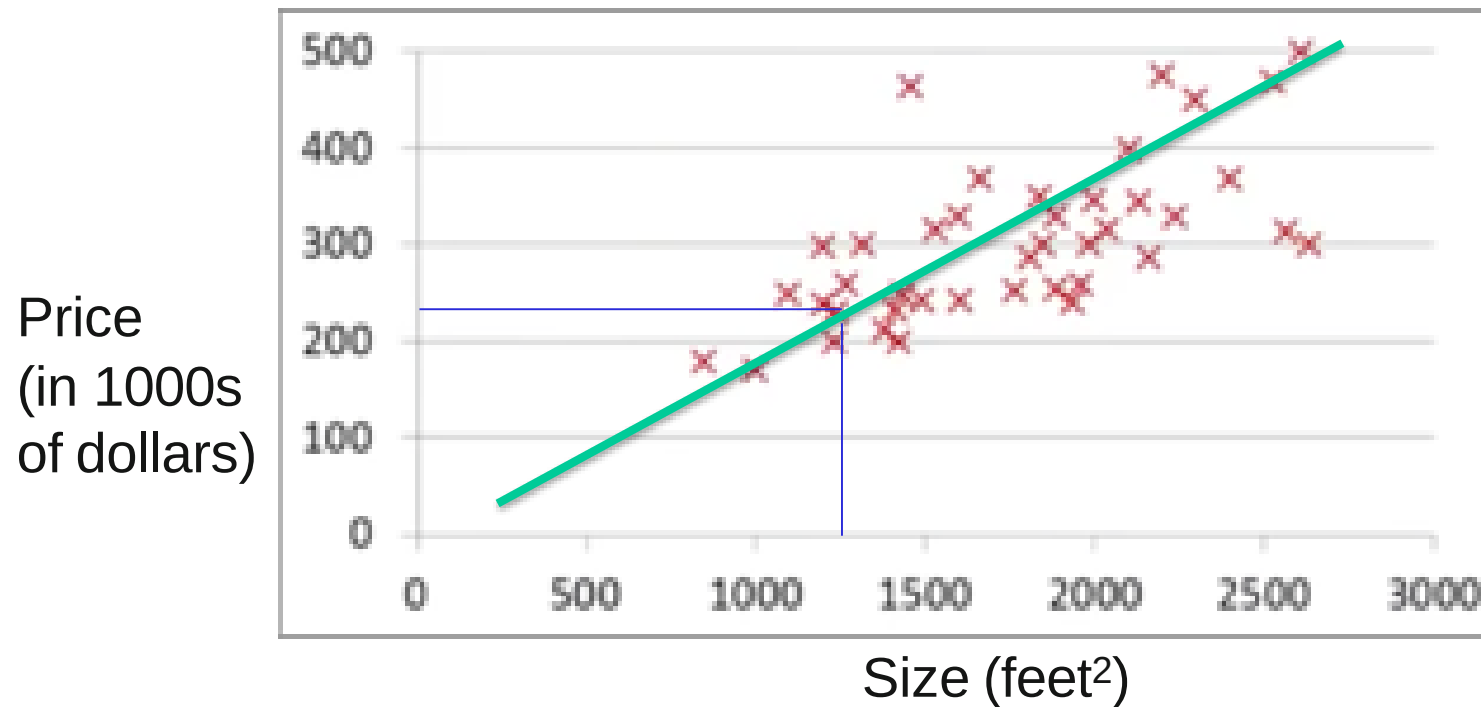
How much a house of 1250ft<sup>2</sup> costs?

We may assume a linear curve,

So, conclude that a house with 1250ft<sup>2</sup> costs 230K\$.

# Motivation

Given the following **housing prices**



## Supervised Learning:

Given the right answers for each example in the data (training data)

## Regression Problem:

Predict real-valued output

Remember that classification (not regression) refers to predicting discrete-valued output

# Motivation

Given the following **training set** of housing prices:

Our job is to learn  
from this data how  
to predict prices

| Size in feet <sup>2</sup> (x) | Price (\$) in 1000s (y) |
|-------------------------------|-------------------------|
| 2104                          | 460                     |
| 1416                          | 232                     |
| 1534                          | 315                     |
| 852                           | 178                     |
| ...                           | ...                     |

## Notation:

$m$  = Number of training examples

$x$ 's = "input" variable/features

$y$ 's = output variable/target variable

$(x, y)$  : a training example

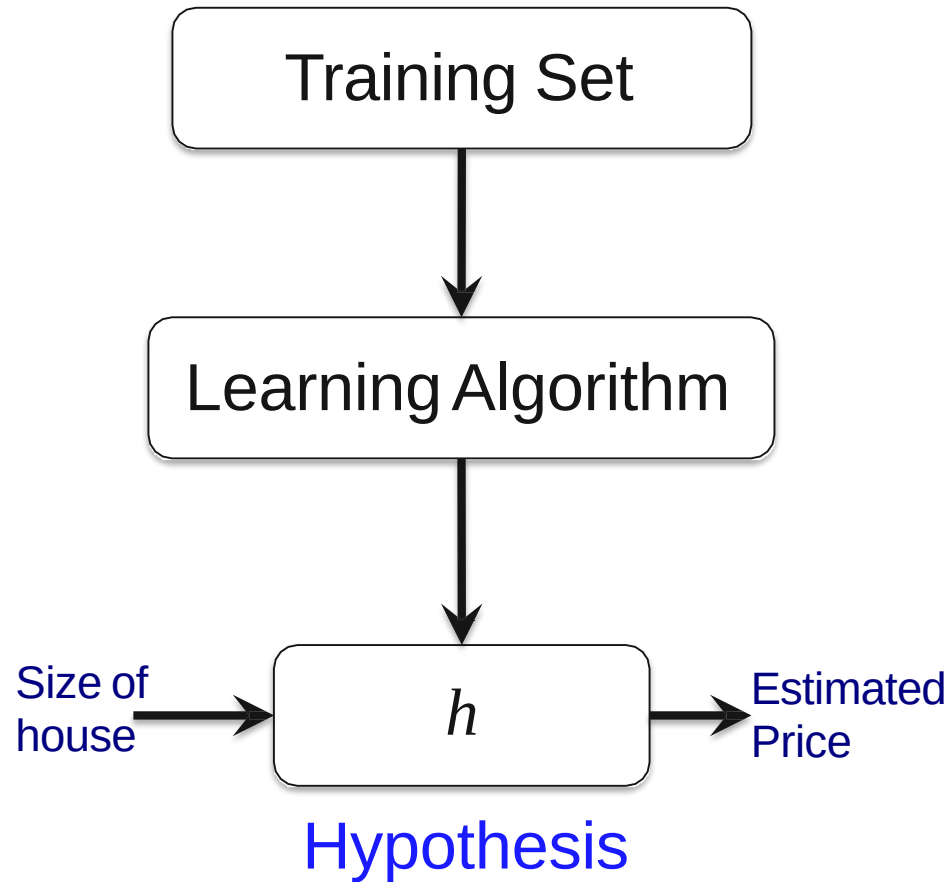
$(x^i, y^i)$  : the  $i^{\text{th}}$  training example

For example:

$x^1 = 2104$

$y^1 = 460$

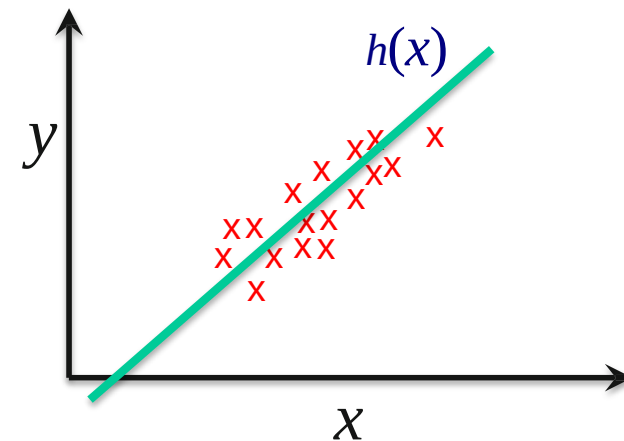
# Univariate Linear Regression



$h$  maps  $x$ 's to  $y$ 's

How to represent  $h$ ?

$$h(x) = \theta_0 + \theta_1 x$$



- This is a linear function.
- Also called linear regression with one variable.

# Multivariate Linear Regression (Multiple Features)

Linear regression with multiple features is also called  
*Multivariate / multiple linear regression*

Suppose we have the following features

| $x_1$                | $x_2$    | $x_3$  | $x_4$ | $y$   |
|----------------------|----------|--------|-------|-------|
| Size ft <sup>2</sup> | bedrooms | floors | Age   | Price |
| 2104                 | 5        | 1      | 45    | 460   |
| 1416                 | 3        | 2      | 40    | 232   |
| 1534                 | 3        | 2      | 30    | 315   |
| 852                  | 2        | 1      | 36    | 178   |
| ...                  | ...      | ...    | ...   | ...   |

A hypothesis function  $h(x)$  might be:

$$h(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \dots + \theta_n x_n$$

or 
$$h(x) = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \dots + \theta_n x_n \quad (x_0=1)$$

e.g., 
$$h(x) = 80 + 0.1 \cdot x_1 + 0.01 \cdot x_2 + 3 \cdot x_3 - 2 \cdot x_4$$

# Polynomial Linear Regression

Suppose we have the following features

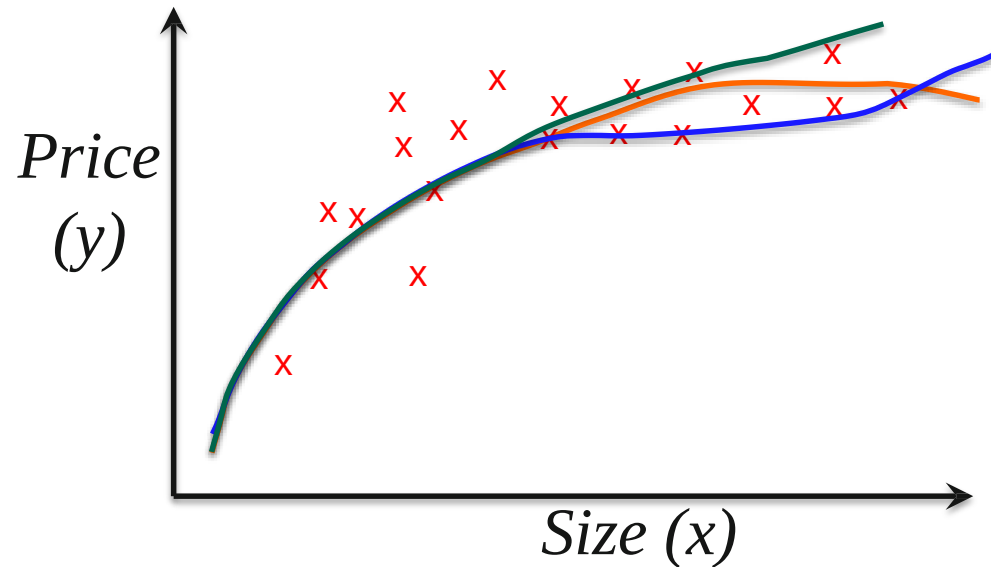
| $x_1$                | $x_2$    | $x_3$  | $x_4$ | $y$   |
|----------------------|----------|--------|-------|-------|
| Size ft <sup>2</sup> | bedrooms | floors | Age   | Price |
| 2104                 | 5        | 1      | 45    | 460   |
| 1416                 | 3        | 2      | 40    | 232   |
| 1534                 | 3        | 2      | 30    | 315   |
| 852                  | 2        | 1      | 36    | 178   |
| ...                  | ...      | ...    | ...   | ...   |

Another hypothesis function  $h(x)$  might be (*polynomial*):

$$h(x) = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2^2 + \theta_3 x_3^3 + \dots + \theta_n x_n^n \quad (x_0=1)$$



# Polynomial Linear Regression



$$h(x) = \theta_0 + \theta_1 x + \theta_2 x^2$$

$$h(x) = \theta_0 + \theta_1 x + \theta_2 \sqrt{x}$$

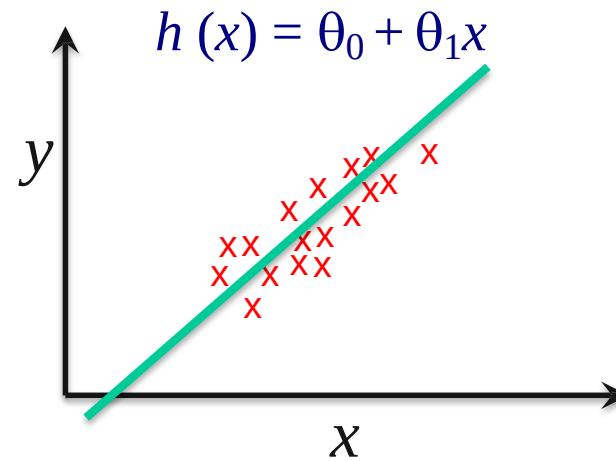
$$h(x) = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3$$

- We may combine features (e.g., size = width \* depth).
- We have the option of what features and what models (quadratic, cubic,...) to use.
- Deciding which features and models that best fit our data and application, is beyond the scope of this course, but there are several algorithms for this.

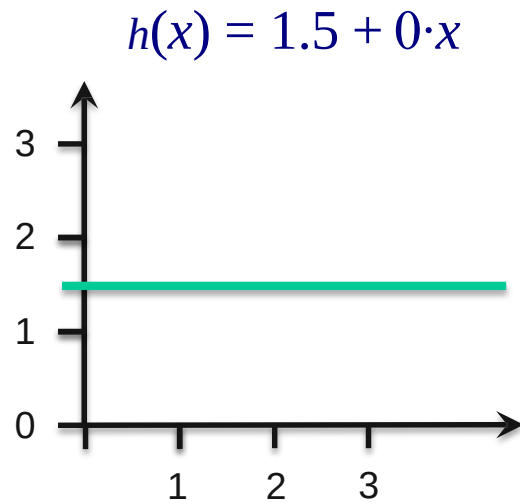
# Understanding $\theta$ s Values

$\theta_i$ 's: Parameters

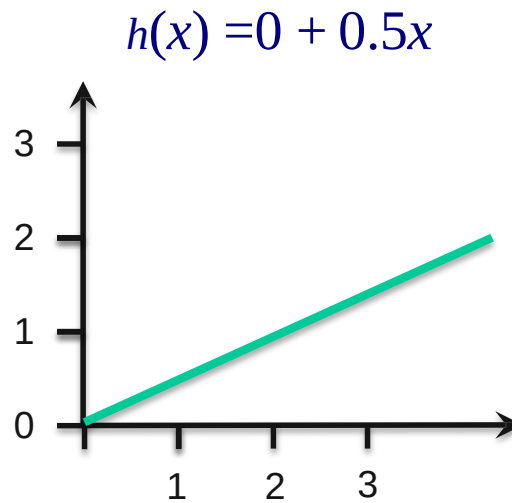
How to choose  $\theta_i$ 's?



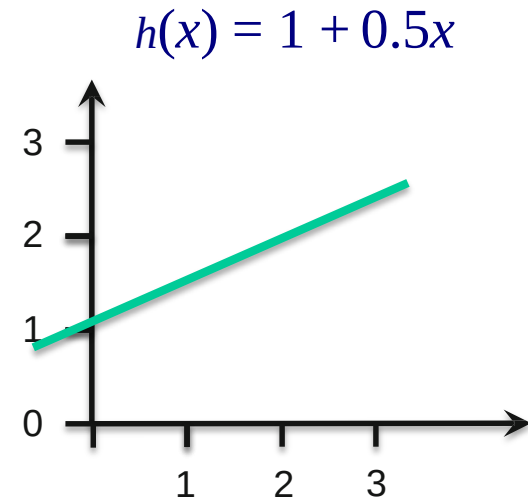
# Understanding $\theta$ s Values



$$\theta_0 = 1.5$$
$$\theta_1 = 0$$

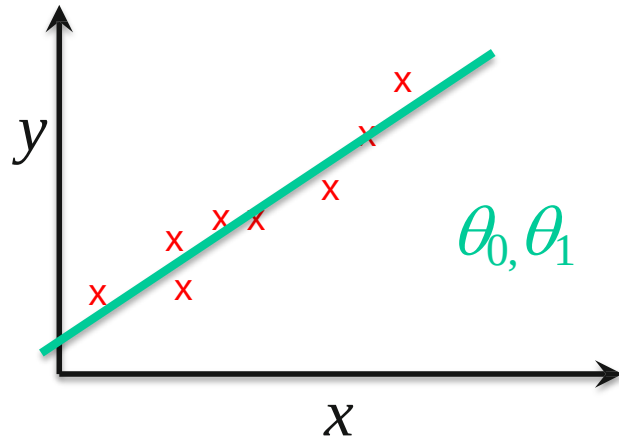


$$\theta_0 = 0$$
$$\theta_1 = 0.5$$



$$\theta_0 = 1$$
$$\theta_1 = 0.5$$

# The Cost Function



Idea: Choose  $\theta_0, \theta_1$  so that  $h(x)$  is close to  $y$  for our training examples  $(x, y)$ .

$y$ : is the actual value.

$h(x)$ : is the estimated value.

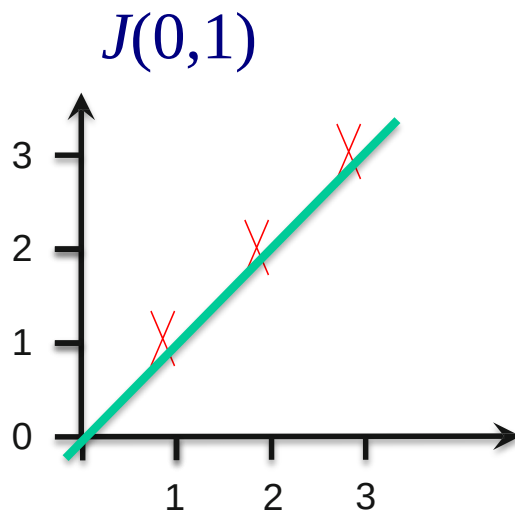
The **Cost Function**  $J$  aims to find  $\theta_0, \theta_1$  that minimizes the error.

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h(x^i) - y^i)^2$$

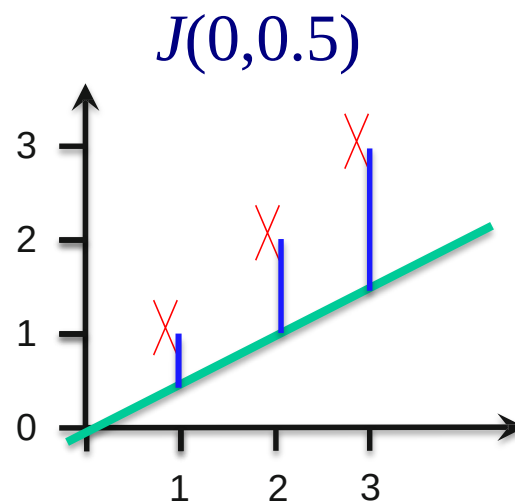
This cost function is also called a **Squared Error function**

# Undersetting the Cost Function

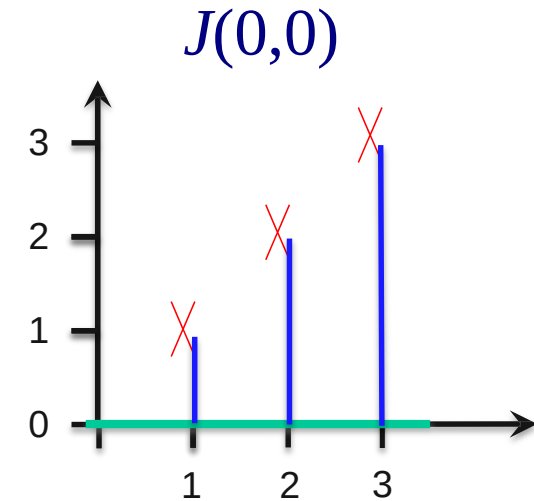
Lets try different values of  $\theta$  and see how the cost function  $J(\theta_0, \theta_1)$  behave!



$$J(1) = 1/2 \cdot 3 [(0)^2 + (0)^2 + (0)^2] = 0$$

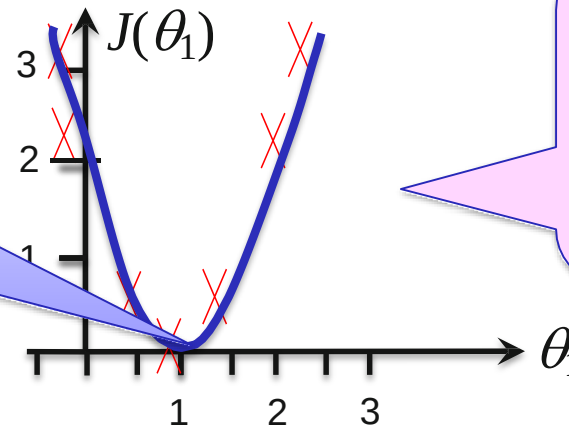


$$J(0.5) = 1/2 \cdot 3 [(0.5-1)^2 + (1-2)^2 + (1.5-3)^2] \simeq 0.68$$



$$J(0) = 1/2 \cdot 3 [(1)^2 + (2)^2 + (3)^2] \simeq 2.3$$

Given our training data, this is the best value of  $\theta_1$ .



But this figure plots only  $\theta_1$ , the problem becomes complicated if we have also  $(\theta_0, \theta_1)$

# Cost Function Intuition II

Remember that our goal is find the minimum values of  $\theta_0$  and  $\theta_1$

**Hypothesis:**  $h(x) = \theta_0 + \theta_1 x_1$

**Parameters:**  $\theta_0, \theta_1$

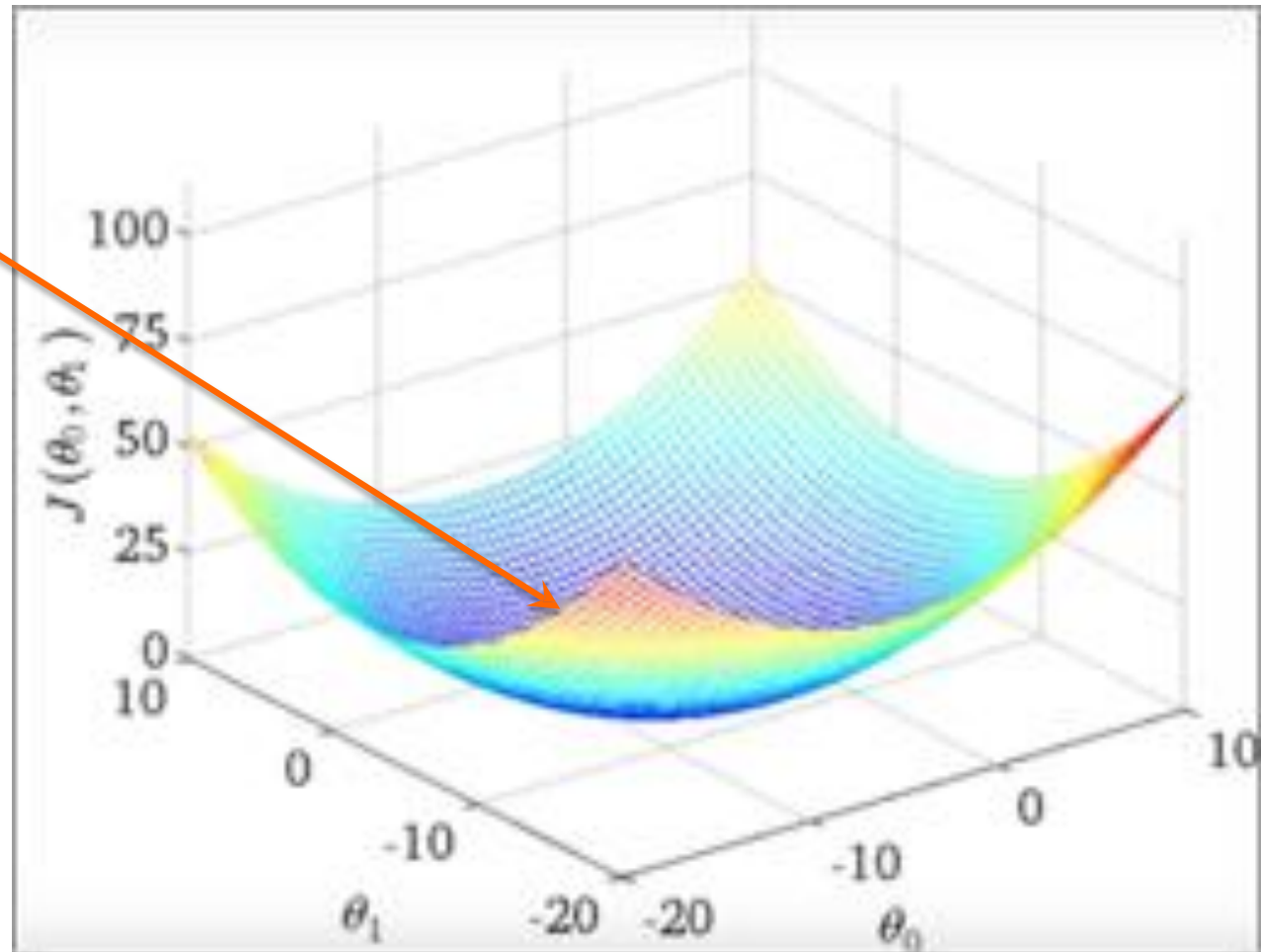
**Cost Function:**  $J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h(x^i) - y^i)^2$

**Our Goal:** minimize  $J(\theta_0, \theta_1)$

## Cost Function Intuition II

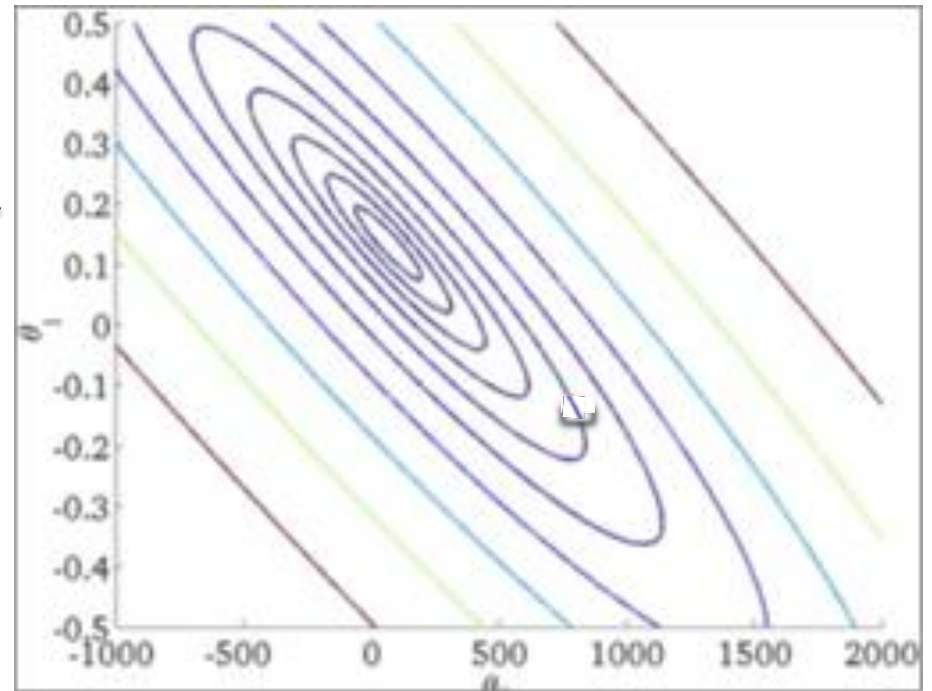
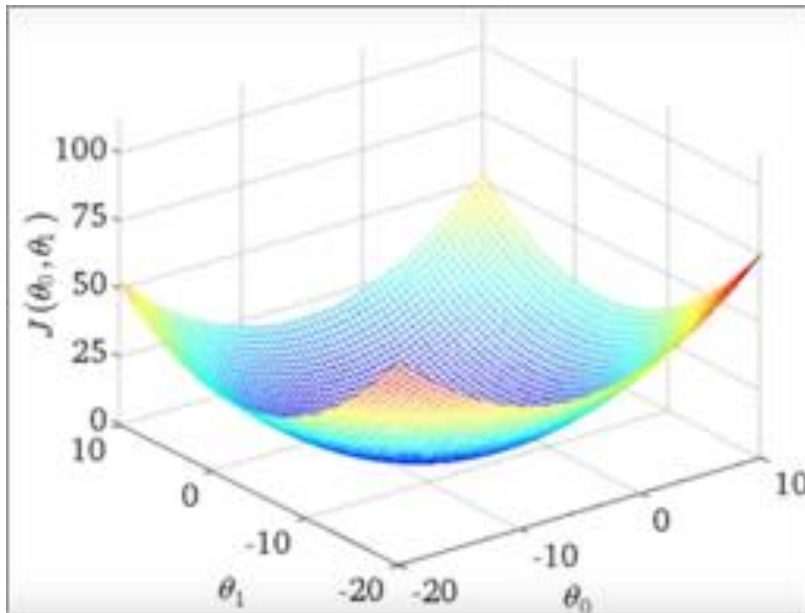
Given any dataset, when we try to draw the cost function  $J(\theta_0, \theta_1)$ , we may get this 3D shape:

Given a dataset,  
our goal is find the  
minimum values of  
 $J(\theta_0, \theta_1)$



## Cost Function Intuition II

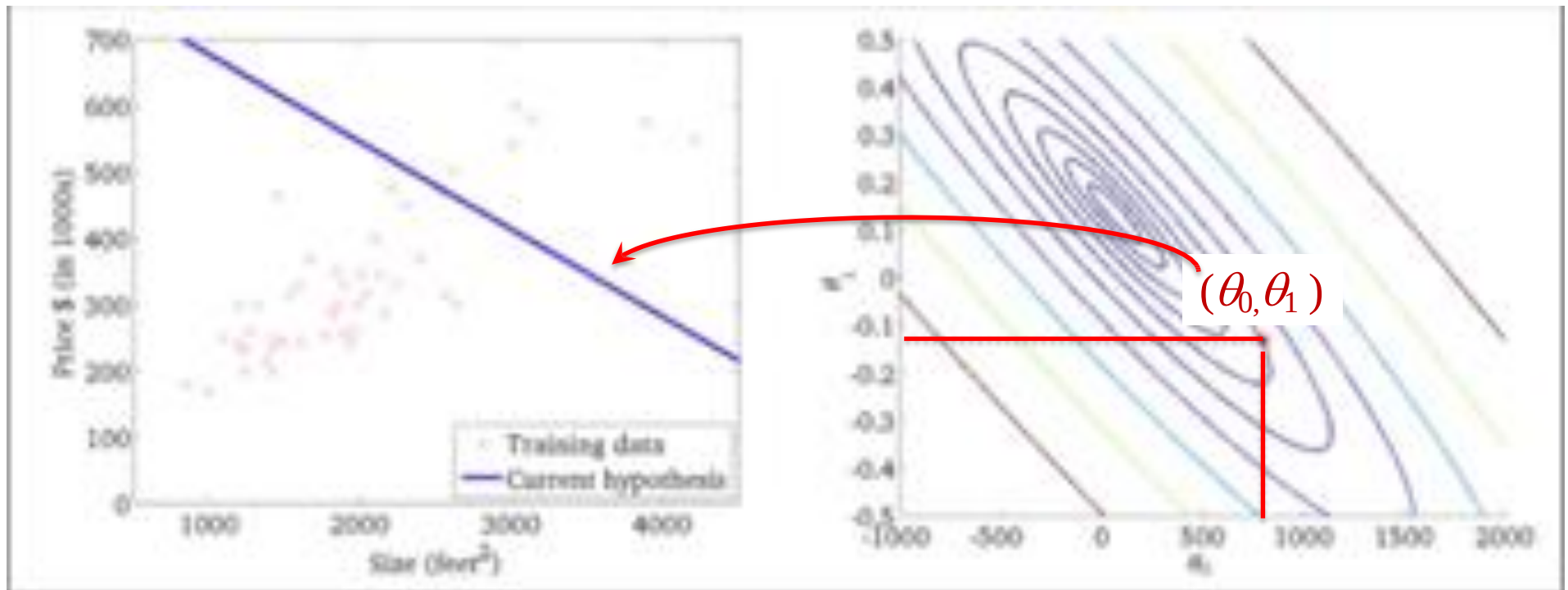
We may draw the cost function also using contour figures:





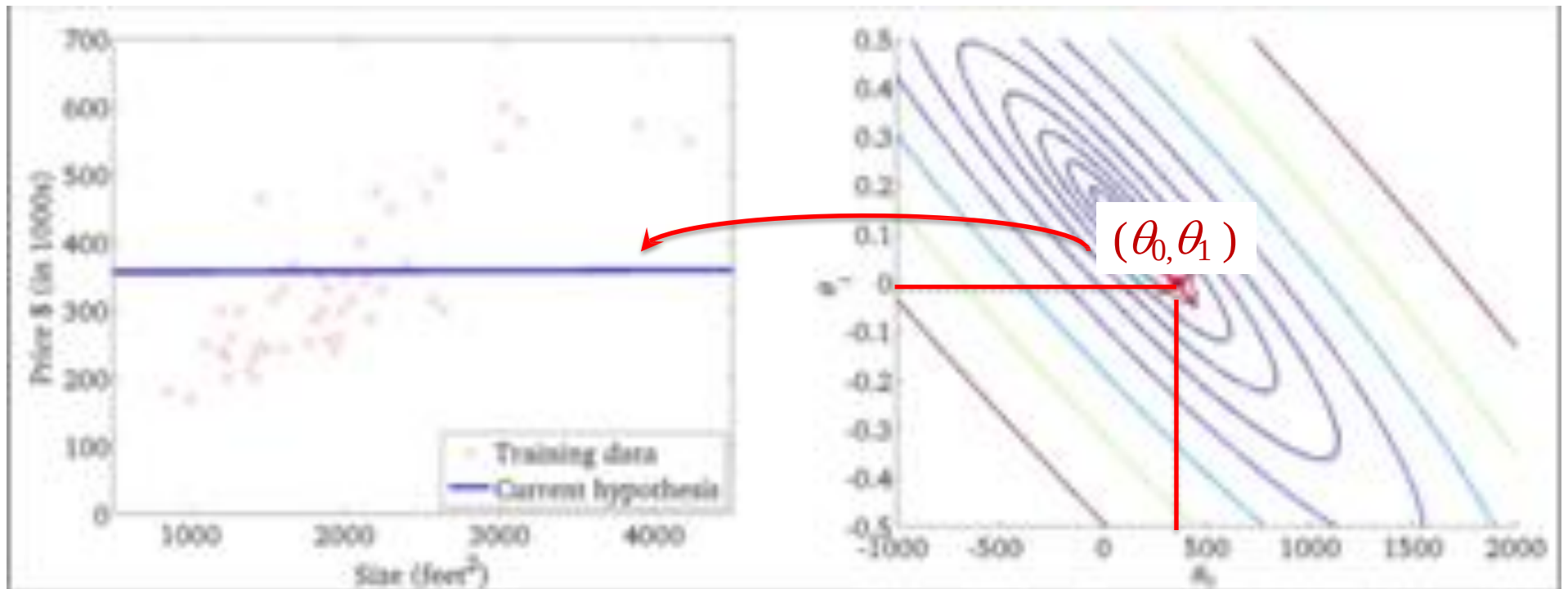
## Cost Function Intuition II

$$h(x) = 800 + 0.15 \cdot x$$

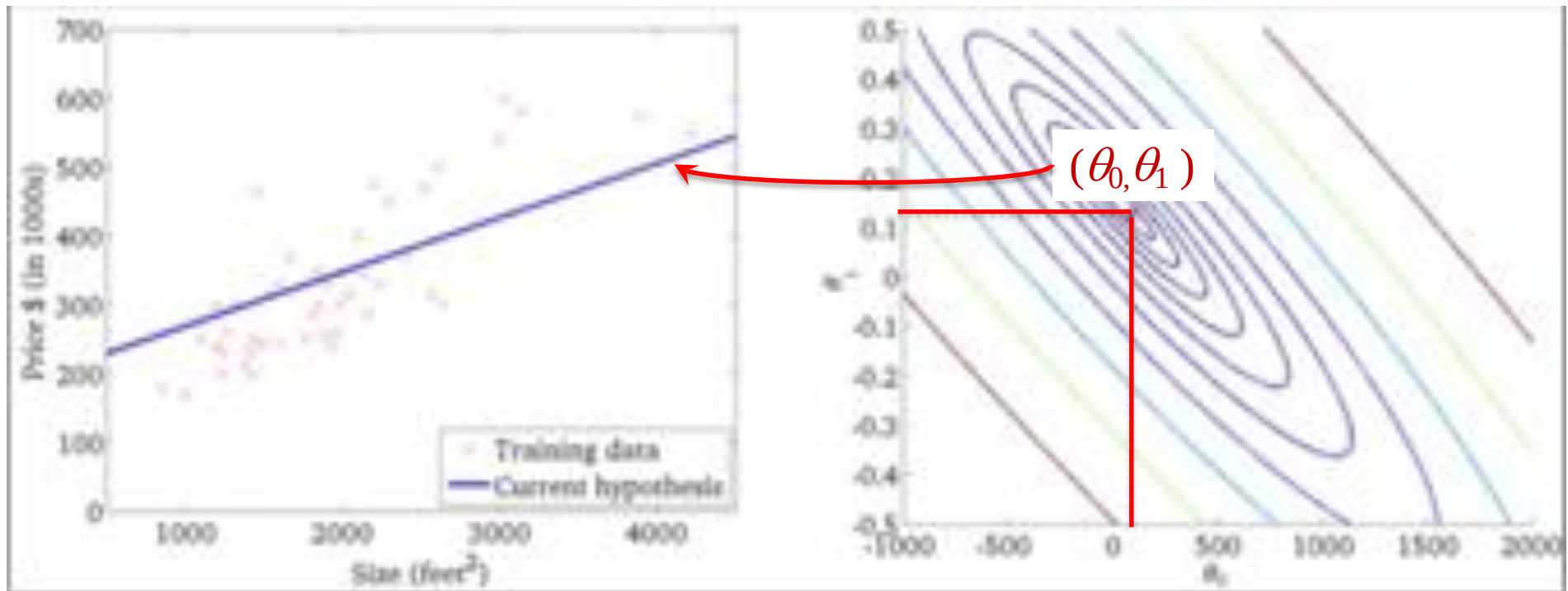


## Cost Function Intuition II

$$h(x) = 360 + 0 \cdot x$$



## Cost Function Intuition II



Is there any way/algorithm to find  $\theta$  automatically?

Yes, e.g., the **Gradient Descent** Algorithm

# The **Gradient Descent** Algorithm

Starts with some initial values of  $\theta_0$  and  $\theta_1$

Keep changing  $\theta_0$  and  $\theta_1$  to reduce ( $\theta_0, \theta_1$ )

Until hopefully we end up at a minimum

$J$  is 0 or 1

Repeat until convergence {

$J$  is 0 or 1

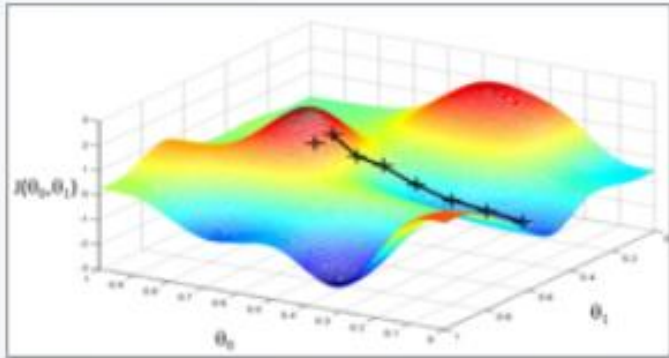
Repeat until convergence {

$$Temp0 := \theta_0 - \alpha \cdot \frac{\partial}{\partial \theta_0} \cdot J(\theta_0, \theta_1)$$
$$Temp1 := \theta_1 - \alpha \cdot \frac{\partial}{\partial \theta_1} \cdot J(\theta_0, \theta_1)$$

---

$$\theta_0 := Temp0$$
$$\theta_1 := Temp1$$

}

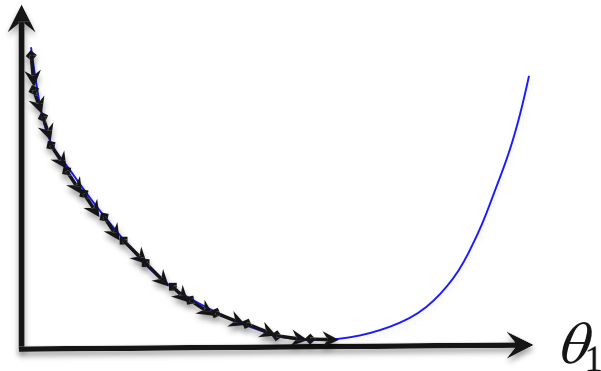


A 3D surface plot of a cost function  $J(\theta_0, \theta_1)$ . The surface is colored with a gradient from blue (low values) to red (high values). A black line with arrows indicates the path of a gradient descent algorithm, starting from a high-cost region and moving towards a local minimum.

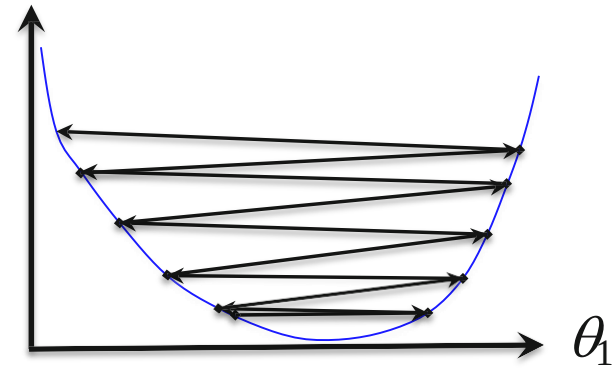
Learning  
Rate

Partial  
Derivation

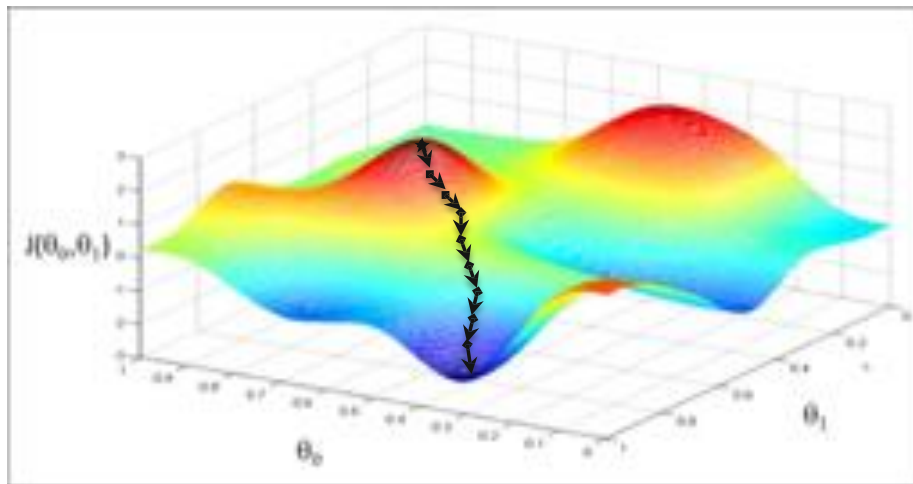
# The Problem of the Gradient Descent algo.



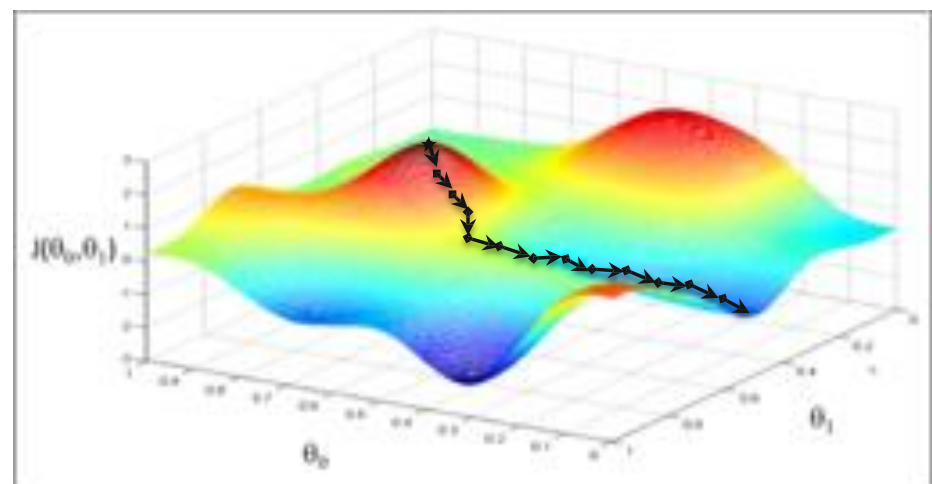
If  $\alpha$  is too small, gradient descent can be slow



If  $\alpha$  is too large, gradient descent can overshoot the minimum. It may fail to converge or even diverge



May converge to *global* minimum



May converge to a *local* minimum

# The Gradient Descent for Linear Regression

Simplified version for linear regression

*J* is 0 and 1

Repeat until convergence {

$$\theta_0 := \theta_0 - \alpha \cdot \frac{1}{m} \cdot \sum_{i=1}^m (h(x^i) - y^i)$$

$$\theta_1 := \theta_1 - \alpha \cdot \frac{1}{m} \cdot \sum_{i=1}^m (h(x^i) - y^i) \cdot x^i$$

}